

Ziyu Xu

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Education

Carnegie Mellon University

Master of Science in Robotics (MSR)

August 2026 – Present

University of Michigan, Ann Arbor

Bachelor of Science in Mechanical Engineering (Robotics Track); Minor in Computer Science

October 2024 – May 2026

GPA: 3.86/4.0

Shanghai Jiao Tong University (UM-SJTU Joint Institute)

Bachelor of Science in Electrical and Computer Engineering

September 2022 – July 2026

GPA: 3.56/4.0

Research Experiences

Robotic Perception and Control, University of Michigan, Ann Arbor

June 2025 – Present

Research Assistant, CURLY Lab, advised by Prof. Maani Ghaffari

Ann Arbor, MI

- **Learning-Augmented Control for Robotics:** Developed closed-loop control pipelines for robotics systems under disturbance, implementing MPC and Lie-MPC with real-time learning-based disturbance estimation.
- **Sim-to-Real Robotic Deployment:** Transferred the autonomy stack from simulation to a commercial surface vehicle platform for algae collection, resolving discrepancies in sensing and dynamics to support robust real-world operation.
- **Robotic Perception:** Built a DINOv3 + SAM2 segmentation pipeline with customized detector head (mAP@0.5 = 41.3%); fine-tuned Grounded-DINO & YOLO+ SAM2 for multi-frame tracking, improving IDF1 by ~50%.
- **Sensor Calibration & System Integration:** Calibrated LiDAR-IMU-camera extrinsics and timing and streamlined the full deployment stack, enabling reliable single-operator experimentation.

Robotic Perception and Algorithm Optimization, University of Michigan, Ann Arbor

October 2024 – Present

Research Assistant, IRAL Lab, advised by Prof. Vasileios Tzoumas

Ann Arbor, MI

- **Multi-Robot Simulation & Dynamics for Autonomous Systems:** Rebuilt a Unity + AirSim + ROS simulation stack for autonomous robotics research, creating a reproducible multi-platform workflow for experimentation.
- **Resource-Aware Coordination / Active SLAM:** Implemented SlideSLAM in ROS1 and initiated a ROS2 port; integrated with AirSim for mapping; established bandwidth-limited test scenarios to evaluate coordination policies
- **Distributed Decision-Making & Submodular Optimization:** Studied submodular maximization and EXP3-based greedy schemes; proposed algorithmic improvements with theoretical validation and a simulation plan

Surgical Computer Vision, Shanghai Jiao Tong University

April 2024 – December 2024

Research Assistant, advised by Prof. Yutong Ban

Shanghai, China

- **ML and Computer Vision:** Developed learning-based perception models for surgical scene understanding using CNN architectures, including ResNet and UNet, on the CholecTriplet dataset. Designed a hybrid spatio-temporal attention mechanism that improved mIoU by 4.8% over prior methods.

Publications

1. Yinan Dong, **Ziyu Xu**, Tsimafei Lazouski, Sangli Teng, and Maani Ghaffari.
“**Online Learning-Enhanced Lie Algebraic MPC for Robust Autonomous Surface Vehicle Control.**”
IEEE Robotics and Automation Letters (RA-L), submitted, 2025.
2. Yutong Ban, **Ziyu Xu**, et al.,
“**Systematic Evaluation and Guidelines for Segment Anything Model in Surgical Video Analysis.**”
npj Digital Surgery, **accepted**, 2024.

Awards

- M Prize in 2024 Mathematical Contest in Modeling (top 13%)
- Gold Medal in 2023 University Physics Competition, by American Physical Society & American Astronomical Society (top 2%)
- First place in 2023 RobMaster Competition, International (top 1%)
- Dean's List and University Honors, University of Michigan, 2024–2025

Skills

- Robotics: ROS/ROS2, control, sensor calibration, simulation, real-world system integration
- Learning & Perception: PyTorch, OpenCV, deep learning, computer vision, segmentation, tracking
- Systems: Python, C++, Linux, Docker, Git, Unity, AirSim
- Mathematical Tools: optimization, probability, dynamics, submodular optimization, CAD, FEA